



《数值分析》 13

主要内容:

切比雪夫插值结点

埃尔米特插值函数

分段插值函数

拉格朗日插值余项

$$R_n(x) = f(x) - L_n(x) = \frac{f^{(n+1)}(\xi_n)}{(n+1)!} \omega_{n+1}(x)$$

其中, $\omega_{n+1}(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$

目标/做什么： 如何选取 $x_0, x_1, \cdots, x_n$ 更好?
(这里：切比雪夫多项式)

切比雪夫插值结点

$f(x) \in C[-1, 1]$, 令 $x = \cos \theta$, 则有 $[-1, 1] \leftrightarrow [0, \pi]$

将 $g(\theta) = f(\cos \theta)$ 展开成余弦级数

$$g(\theta) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos n\theta$$

百度百科中: 正弦级数和余弦级数

n 阶切比雪夫多项式: $T_n = \cos(n\theta)$

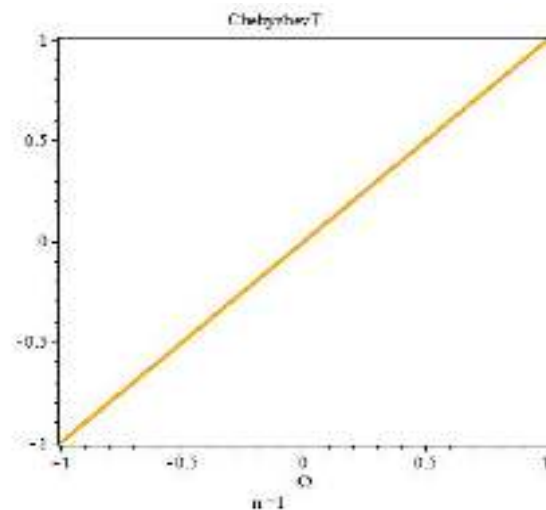
$$\cos(n+1)\theta = 2\cos n\theta \cos \theta - \cos(n-1)\theta$$

$$T_{n+1} = 2xT_n - T_{n-1} \quad (n = 1, 2, \dots)$$

$$T_0 = 1 \quad T_1 = x$$

$$T_2 = 2x^2 - 1 \quad \rightarrow \quad x_0^{(2)} = -\frac{1}{\sqrt{2}}, \quad x_1^{(2)} = \frac{1}{\sqrt{2}}$$

$$T_3 = 4x^3 - 3x \quad \rightarrow \quad x_0^{(3)} = -\frac{\sqrt{3}}{2}, \quad x_1^{(3)} = 0, \quad x_2^{(3)} = \frac{\sqrt{3}}{2}$$



切比雪夫插值结点

$$\cos n\theta = 0 \rightarrow n\theta = \frac{(2k+1)\pi}{2} \rightarrow \theta = \frac{(2k+1)\pi}{2n}$$

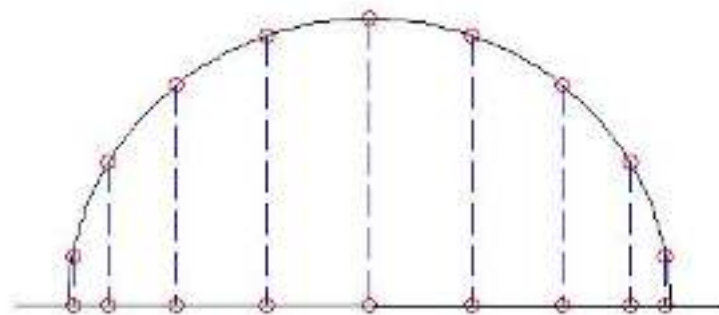
$$\cos \theta = x \Leftrightarrow \theta = \arccos x$$

$$\rightarrow \arccos x = \frac{(2k+1)\pi}{2n} \rightarrow x_k = \cos\left(\frac{(2k+1)\pi}{2n}\right)$$

$$(k=0,1,\cdots,n-1)$$

n 次多项式插值的切比雪夫结点

$$\rightarrow x_k = \cos\left(\frac{(2k+1)\pi}{2(n+1)}\right) \quad (k=0,1,\cdots,n)$$



切比雪夫插值结点

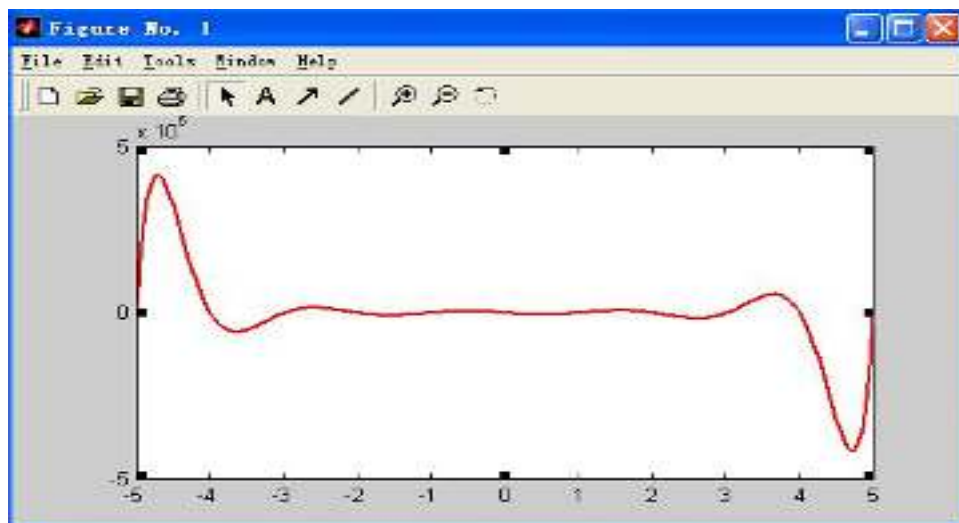
例1. 函数 $f(x) = \frac{1}{1+x^2} \quad x \in [-5, 5]$

取等距插值结点: $-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$

$$f(x) = L_{10}(x) + \frac{f^{(11)}(\xi_n)}{11!} \omega_{11}(x)$$

$$\omega_{11}(x) = (x+5)(x+4)(x+3)(x+2)(x+1)x(x-1)(x-2)(x-3)(x-4)(x-5)$$

$\omega_{11}(x) \rightarrow$



切比雪夫插值结点

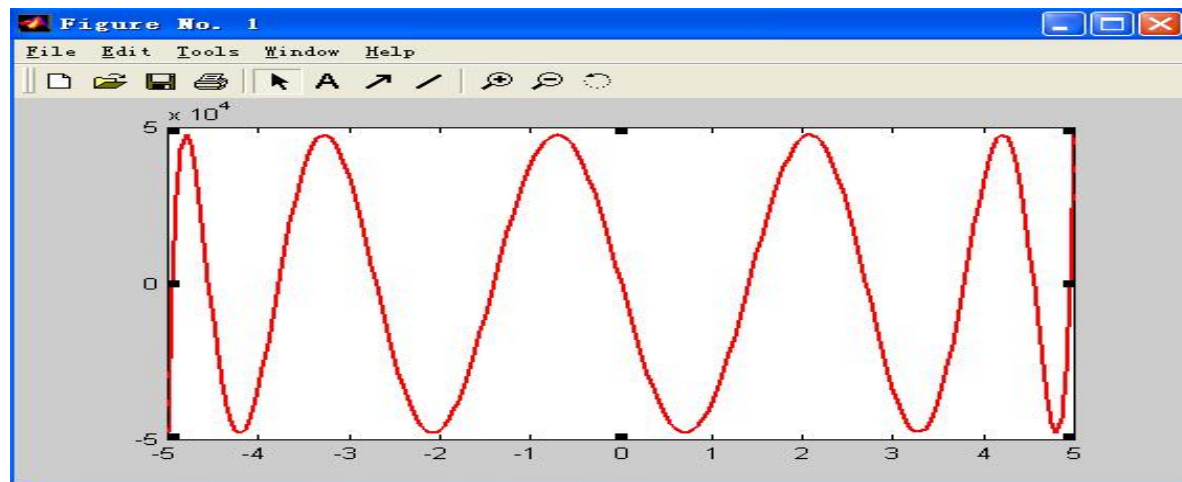
在 $[-5, 5]$ 区间上,取11个切比雪夫结点

$$x_k = 5 \cos\left(\frac{(2k+1)\pi}{22}\right) \quad (k=10, 9, 8, \dots, 1, 0)$$

-4.9491	-4.5482	-3.7787	-2.7032	-1.4087	0.0000	1.4087
2.7032	3.7787	4.5482	4.9491			

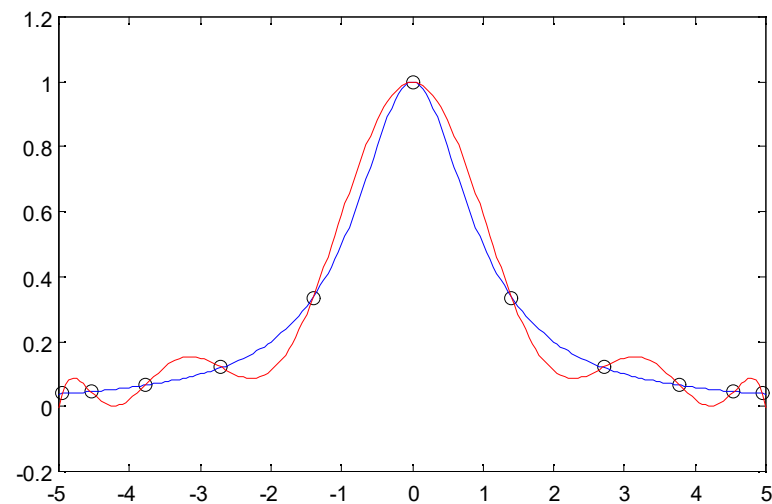
$$\omega_{11}(x) = (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{10})$$

$\omega_{11}(x) \rightarrow$

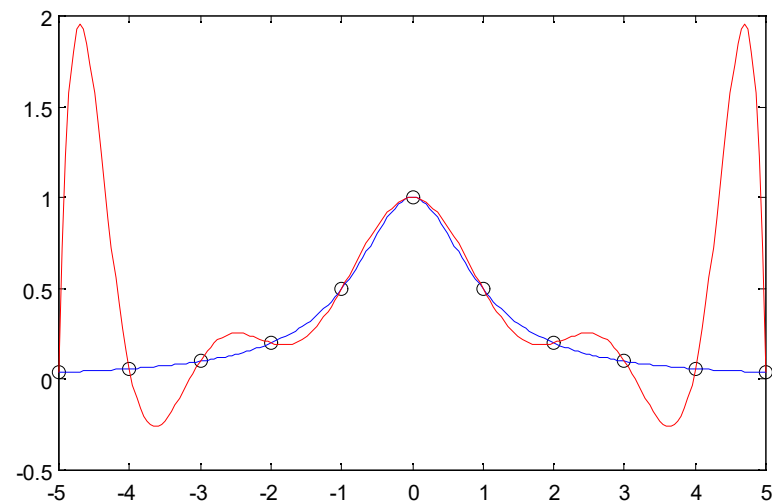


切比雪夫插值结点

插值函数 $L_{10}(x)$ 取
切比雪夫结点插值



插值函数 $L_{10}(x)$ 取
等距结点插值



插值条件中除函数值插值条件外，还有导数值插值条件，

即：

已知 $2n+2$ 个条件

x_i	x_0	x_1	$\dots$	x_n
$y_i = f(x_i)$	y_0	y_1	$\dots$	y_n
$y'_i = f'(x_i)$	y'_0	y'_1	$\dots$	y'_n

求：一个次数不超过 $2n+1$ 的多项式 $H_{2n+1}(x)$

三次Hermite插值问题 设 $f(x) \in C^4[x_0, x_1]$

已知在插值节点 x_0 和 x_1 的函数值和导数值为:

$$\begin{aligned} f(x_0) &= y_0 & f(x_1) &= y_1 \\ f'(x_0) &= m_0 & f'(x_1) &= m_1 \end{aligned}$$

可以求到次数为3次的多项式 $H_3(x)$, 称为 **三次Hermite 插值多项式**

插值条件:

$$\begin{aligned} H(x_0) &= y_0 & H(x_1) &= y_1 \\ H'(x_0) &= m_0 & H'(x_1) &= m_1 \end{aligned}$$

埃尔米特插值

采用基函数方式构造 $H(x)$:

$$H(x) = \alpha_0(x)y_0 + \alpha_1(x)y_1 + \beta_0(x)m_0 + \beta_1(x)m_1$$

插值条件:

$$\begin{aligned} H(x_0) &= y_0 & H(x_1) &= y_1 \\ H'(x_0) &= m_0 & H'(x_1) &= m_1 \end{aligned}$$

插值条件表

	函数值		导数值	
	x_0	x_1	x_0	x_1
$\alpha_0(x)$	1	0	0	0
$\alpha_1(x)$	0	1	0	0
$\beta_0(x)$	0	0	1	0
$\beta_1(x)$	0	0	0	1

如何求 $\alpha_0(x)$ $\alpha_1(x)$ $\beta_0(x)$ $\beta_1(x)$

$\alpha_0(x)$

由于 $\alpha_0(x_1) = \alpha'_0(x_1) = 0$, 故 $\alpha_0(x)$ 含有 $(x - x_1)^2$ 因子。可设

$$\alpha_0(x) = (a + b(x - x_0))(x - x_1)^2$$

其中 a, b 为待定系数。

$$\text{由 } \alpha_0(x_0) = 1, \text{ 可得 } a = \frac{1}{(x_0 - x_1)^2}.$$

$$\text{由 } \alpha'_0(x_0) = 0, \text{ 可得 } b = \frac{-2}{(x_0 - x_1)^3}.$$

将 a, b 代入得

$$\alpha_0(x) = \left(1 + 2 \frac{x - x_0}{x_1 - x_0} \right) \left(\frac{x - x_1}{x_0 - x_1} \right)^2$$

$$\alpha_0(x) = \left(1 + 2 \frac{x - x_0}{x_1 - x_0}\right) \left(\frac{x - x_1}{x_0 - x_1}\right)^2$$

类似地，将 x_0, x_1 互换，可得

$$\alpha_1(x) = \left(1 + 2 \frac{x - x_1}{x_0 - x_1}\right) \left(\frac{x - x_0}{x_1 - x_0}\right)^2$$

$$\beta_0(x)$$

由于 $\beta_0(x_0) = \beta_0(x_1) = \beta'_0(x_1) = 0$, 故 $\beta_0(x)$ 含有 $(x-x_0)(x-x_1)^2$

因子。可设

$$\beta_0(x) = c(x-x_0)(x-x_1)^2$$

其中 c 为待定系数。

由 $\beta'_0(x_0) = 1$, 可得 $c = \frac{1}{(x_0 - x_1)^2}$.

$$\beta_0(x) = (x-x_0) \left(\frac{x-x_1}{x_0-x_1} \right)^2$$

类似地, 将 x_0, x_1 互换, 可得

$$\beta_1(x) = (x-x_1) \left(\frac{x-x_0}{x_1-x_0} \right)^2$$

最终求得所有4个基函数（针对三次Hermite插值）

$$\begin{aligned}\alpha_0(x) &= \left(1 + 2 \frac{x - x_0}{x_1 - x_0}\right) \left(\frac{x - x_1}{x_0 - x_1}\right) & 2\beta_0(x) &= (x - x_0) \left(\frac{x - x_1}{x_0 - x_1}\right)^2 \\ \alpha_1(x) &= \left(1 + 2 \frac{x - x_1}{x_0 - x_1}\right) \left(\frac{x - x_0}{x_1 - x_0}\right)^2 & \beta_1(x) &= (x - x_1) \left(\frac{x - x_0}{x_1 - x_0}\right)^2\end{aligned}$$

代入4个基函数即可得：三次Hermite插值多项式

$$H(x) = \alpha_0(x)y_0 + \alpha_1(x)y_1 + \beta_0(x)m_0 + \beta_1(x)m_1$$

定理： 两点三次Hermite插值的误差估计式

$$R(x) = f(x) - H_3(x) = \frac{f^{(4)}(\xi)}{4!} [(x - x_0)(x - x_1)]^2$$

证明：由插值条件知

$$R(x_0) = R'(x_0) = 0 \quad R(x_1) = R'(x_1) = 0$$

取 x 异于 x_0 和 x_1 , 有

$$R(x) = C(x)(x - x_0)^2(x - x_1)^2$$

利用 $f(x) - H(x) = C(x)(x - x_0)^2(x - x_1)^2$

构造辅助函数

$$F(t) = f(t) - H(t) - C(x)(t - x_0)^2(t - x_1)^2$$

显然, $F(t)$ 有三个零点 x_0, x, x_1 , 由Roll定理知, 存在 $F'(t)$ 两个零点 t_0, t_1 . 故 $F'(t)$ 有四个相异零点

$$x_0 < t_0 < x < t_1 < x_1$$

反复应用 Roll 定理, 得 $F^{(4)}(t)$ 知一个零点设为 ξ

$$F(t) = f(t) - H(t) - C(x)(t - x_0)^2(t - x_1)^2$$

$$\rightarrow F^{(4)}(\xi) = f^{(4)}(\xi) - C(x)(4!) = 0$$

$$\rightarrow C(x) = \frac{f^{(4)}(\xi)}{4!}$$

$$R(x) = \frac{f^{(4)}(\xi)}{4!} [(x - x_0)(x - x_1)]^2$$

分段插值

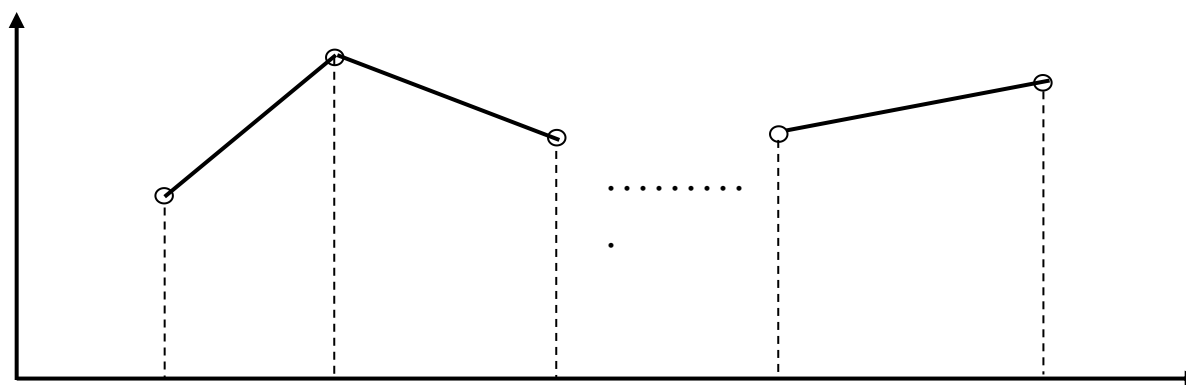
分段线性插值

插值节点满足: $x_0 < x_1 < \cdots < x_n$ 已知

$$y_j = f(x_j) \quad (j = 0, 1, 2, \cdots, n)$$

$x \in [x_j, x_{j+1}]$ 时, 线性插值函数

$$L_h(x) = \frac{x_{j+1} - x}{x_{j+1} - x_j} y_j + \frac{x - x_j}{x_{j+1} - x_j} y_{j+1} \quad (j = 0, 1, \cdots, n-1)$$



分段三次Hermite插值

已知函数值和导数值 $y_j = f(x_j), m_j = f'(x_j)$

$$\begin{aligned} H_h(x) = & \left(1 + 2 \frac{x - x_j}{x_{j+1} - x_j}\right) \left(\frac{x_{j+1} - x}{x_{j+1} - x_j}\right)^2 y_j \quad (j=0,1,2,\cdots,n) \\ & + \left(1 + 2 \frac{x_{j+1} - x}{x_{j+1} - x_j}\right) \left(\frac{x - x_j}{x_{j+1} - x_j}\right)^2 y_{j+1} \\ & + (x - x_j) \left(\frac{x_{j+1} - x}{x_{j+1} - x_j}\right)^2 m_j + (x - x_{j+1}) \left(\frac{x - x_j}{x_{j+1} - x_j}\right)^2 m_{j+1} \end{aligned}$$

$$x \in [x_j, x_{j+1}] \quad (j=0,1,2,\cdots,n-1)$$

切比雪夫插值结点

埃尔米特插值

分段插值函数